

Generative AI Text Summarization using Gemini - Detailed Notes

1. AI Lifecycle

The AI Lifecycle describes the complete process of building and maintaining an AI solution.

Stages:

- Define Problem & Objectives – Identify the problem and expected outcomes.
- Collect & Prepare Data – Gather and clean relevant data.
- Build & Train Model – Train AI/ML models on prepared data.
- Evaluate Model – Measure performance using suitable metrics.
- Deploy Model – Make the model available to users.
- Monitor & Maintain – Track performance after deployment.
- Improve & Iterate – Continuously refine the system.

The lifecycle is iterative and improvements are made continuously based on feedback.

2. Project Objective

The objective of this project is to build a Generative AI powered text summarization system using Google's Gemini model. **Goals:**

- Accept long textual content.
- Generate concise summaries automatically.
- Improve readability and information extraction.
- Evaluate summary quality using AI itself.

3. Installing Required Libraries

Python libraries used in the project: • google-genai / Google Gemini SDK – Access Gemini models.

- pandas – Data manipulation and tabular datasets.
- matplotlib – Visualization and analysis.

Example: `pip install google-genai pandas matplotlib`

4. Google GenAI Client Setup

To use Gemini, an API key must be obtained from Google AI Studio. **Steps:**

1. Create an API key.
2. Store the key securely.
3. Initialize the Gemini client.
4. Verify successful connection.

The client acts as the communication bridge between the application and Gemini models.

5. Creating the Dataset

A dataset is created using pandas DataFrames. Example workflow: • Create a Python dictionary containing text samples.

- Convert the dictionary into a DataFrame.
- Inspect records using `df.head()`.

Benefits: • Structured storage.

- Easy filtering and processing.
- Compatible with AI workflows.

6. Data Cleaning and Preprocessing

Raw text often contains unnecessary spaces, symbols, and formatting inconsistencies. Typical cleaning operations:

- Remove extra whitespace.

- Remove unwanted characters.
- Normalize text formatting.
- Convert text into a consistent format.

Benefits:

- Better prompt quality.

- More accurate summaries.
- Reduced noise in AI outputs.

7. Build & Train Model – Prompt Engineering

Generative AI models are foundation models. Instead of traditional training, performance is improved through prompt engineering. Example Prompt: "Summarize the following text in 3 bullet points." Good prompts should:

- Be specific.

- Mention desired format.
- Define output length.
- Clearly state the task.

Prompt quality directly affects summary quality.

8. Generating Summaries using Gemini

The cleaned text is sent to Gemini along with a prompt. Process:

1. Prepare input text.

2. Construct prompt.
3. Send request to Gemini.
4. Receive generated response.
5. Extract summary text.

Sample Summary:

- AI automates repetitive tasks.

- AI improves decision-making processes.
- AI enables innovative applications across industries.

9. Summary Analysis

After generating summaries, additional analysis can be performed. Important factors:

- Length of summary.

- Number of bullet points.
- Information coverage.
- Relevance to source text.
- Readability score.

This helps determine whether the generated summary meets project requirements.

10. Quality Evaluation using Gemini

Gemini can also be used as an evaluator. Evaluation Criteria:

- Accuracy**

Measures whether the summary correctly represents the original text.

- Completeness**

Checks whether important information is preserved.

- Readability**

Measures clarity, grammar, and ease of understanding. Example Evaluation Prompt: Evaluate the following summary based on: 1. Accuracy 2. Completeness 3. Readability Provide scores and suggestions for improvement.

11. Complete Workflow

1. Install libraries.
2. Define project objective.

3. Configure Gemini API.
4. Create dataset.
5. Clean text data.
6. Design prompt.
7. Generate summaries.
8. Analyze results.
9. Evaluate quality.
10. Improve prompts and repeat.

12. Viva / Exam Questions

1. What is Generative AI?
2. What is text summarization?
3. Why is data cleaning important?
4. What is prompt engineering?
5. What are Gemini models?
6. What is the role of pandas?
7. What is summary evaluation?
8. Explain the AI lifecycle.
9. How does Gemini generate summaries?
10. Differentiate training and prompt engineering.